# Personal Project: Remote Control Car

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Version 2.0

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Preface

Version 1 of this project was me bullshitting around, just looking through, learning, and understanding remote control (RC) cars and regular car components. Ultimately, looking at other designs on MakerWorld, Cults, and other 3D printing sites, I have decided to just salvage a cheap RC car to save money and get more experience with designing, before making bigger investments. With Version 2, I hope to make functioning components and systems in the old RC car. To the reader, I hope you enjoy reading about my learning experience and appreciate the steps I took.

Original RC Model

After deciding to save money and my sanity with not having to deal with designing electrical components (yet), I went with the Adventure Force (1:22) McLaren 765LT RC Car model, it was $9.97 at Walmart and I get a preassembled RC car that has a PCB board, some generic DC motors, and a battery pack. Due to my circuit skills not being the sharpest, I plan to to use these components for the “upgrade”, I plan to add suspension and possibly later the differential to the RC car, to increase the handling of the RC car. I will note that with reusing the components, the vehicle’s speed probably won’t be what I want, but that will for sure be an objective to worry about in future versions.

Fig 1

Plans for Design

Before making designs in SolidWorks and validating them with Ansys, I must explain designs I am considering for the suspension system.

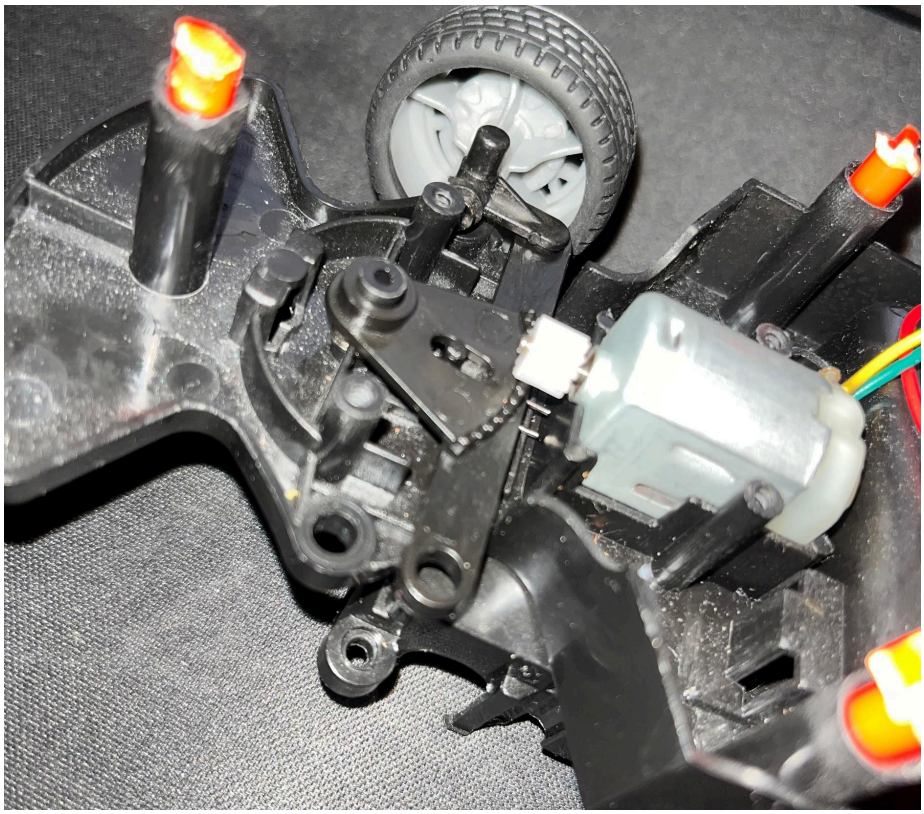

Fig 2

The image above, is the steering located at the front of the car, which has had its chassis removed so I can access all of the components. The two large holes that are next to each other at the bottom of the picture is where one of the wheels are. One of my designs follows a typical suspension design in any RC car and normal sized vehicle, where there is a joint that allows movement for suspension, and the car's track would be extended. Another design I was thinking of was to essentially make the RC car lifted, as in making a shock absorber to go in those two holes. I would like to try the initial design, as that has been proven to work time and again. For the rear suspension, there might be a slight issue, the wheels are directly connected to the shaft. Plans for designing around this is to potentially cut the shaft into three parts, the main shaft and

the two linkages. I am still deciding on ways to connect the metal part with PLA plastic.

(PLEASE PUT A PIC FOR THE REAR SYSTEM)

Modeling and Verifying Design

I was inspired by u/DreamsWhereIamDying modeling for ball and socket joint for the suspension joints

Assembling Design

One particular concern I do have is tolerances

Testing

yap

Failures and What to do Next

Imma probably do a differential on the rear next idk yet tho